

Reading the Room

Baseball's Hidden Poker Game Revealed Through Tracking Data

Team 302

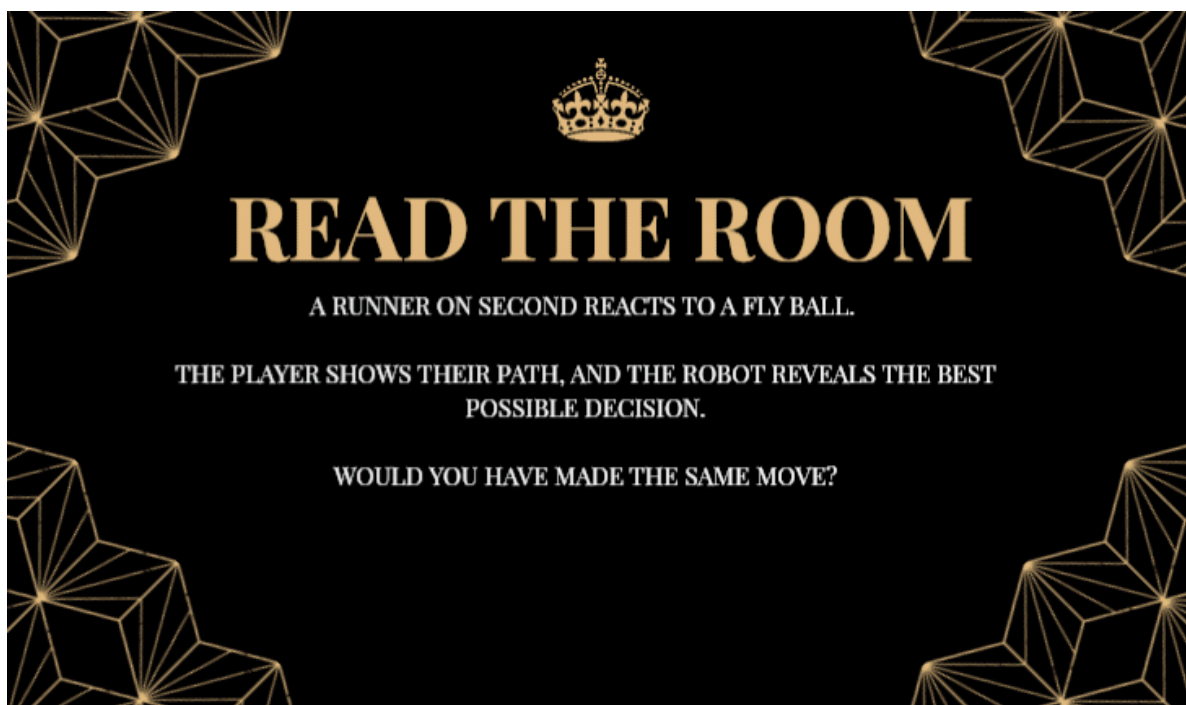
Abstract

Decisions within baseball are often evaluated only after the outcome is known. Fans often judge a baserunner's decision in hindsight, viewing an aggressive advance as brilliant if the ball drops, but disastrous when the fielder makes the catch. As spectators, we are drawn to follow the ball, often overlooking the runner's positioning and the difficult decision being made.

Our project introduces a scoreboard designed to visualize this player decision-making process. Using anonymised player and ball tracking data, we developed and tested models that estimate the probability of a catch, the probability of successfully returning to the base after a catch, and the expected advancement if the ball is not caught. By combining these models, we determine the optimal positioning for a runner throughout the path of a fly ball.

The scoreboard transforms these predictive models into an experience for fans to re-live player decisions. The highlight of the scoreboard is an animation of the runner's actual movement compared to the model's recommended movement. Additionally, a leaderboard ranks players based on how closely their positioning aligns with model recommendations while game information provides context.

When we focus on decision quality, our scoreboard allows everyone from casual fans to baseball professionals to better understand split second decisions from the player's perspective.



[Figure 1:](#) Final Scoreboard interface showcasing baserunner decision quality through rankings, performance summaries, and an interactive play visualization.

(1♥) Playing the Hand You're Dealt: The Baserunner's Dilemma

Baseball often feels like a game of high-stakes poker. Players make decisions with incomplete information, balancing risk and reward. For baserunners, this uncertainty is most evident on fly balls, where a runner on second base must choose between staying close enough to return safely if the catch is made, or moving farther toward home to capitalize if the ball drops. The aggressive move that creates a scoring opportunity could also cost an inning.

This dilemma was recently on display during Game 6 of the 2025 World Series. With the tying run on second base in the bottom of the 9th, Toronto's Addison Barger faced one of the most difficult baserunning decisions a player can make. A fly ball was hit towards left field, carrying into the gap. Barger broke aggressively off of second base, reading the ball off the bat and looking to tie the game as it carried towards the outfield.

For a moment, the ball appeared likely to fall in, creating the possibility of tying the game. Then, in a matter of moments, left fielder Kiké Hernández closed the distance and made the catch. Barger, now too far from second base to recover, was doubled off, forcing Game 7.



Figure 2: World Series play involving Addison Barger, highlighting the baserunning decision that motivated our analysis. Video linked to figure.

To fans who watched after the fact, the decision probably seemed reckless. The result had already been revealed, and Barger's movement became the defining moment of the game. However, baseball decisions are not made with the benefit of hindsight. At the moment Barger moved toward third base, he was uncertain if the ball would be caught. He was weighing the implications of being doubled off with the potential to score the tying run if the ball lands.

Plays like this happen constantly in baseball, yet fans often judge them by the outcome rather than the choice. A runner can be praised as brilliant if the ball drops and criticized after a great play in the field, even when the decision was equally sound.

Our project aims to evaluate these moments before we know the outcome. By modeling the possible outcomes of a fly ball, we estimate the position that maximized a runner's expected value throughout a play.

Our resulting product is a scoreboard designed to reveal the decision behind the outcome to fans. The visualization compares a runner's actual movement to the movement recommended by our model, showing where a player gained or lost value with their positioning. Additionally, a leaderboard ranks runners by how consistently their positioning aligns with model recommendations.

Rather than looking at the outcome of a play, our scoreboard presents a new question to audiences: Did the runner make the right decision before the final card turned?

(2♣) The Scoreboard: Revealing the Hand

The goal of our scoreboard is to make baserunning decisions visible to fans. Rather than only focusing on the outcome of a fly ball, it presents the information needed to understand the runner's decision throughout the play ([Figure 1](#)).

The main attraction of our scoreboard is a play animation that compares the runner's actual movement to the movement recommended by our model. Throughout the flight of the ball, our model evaluates the changing risk and reward at each possible position. This animation lets viewers watch the decision unfold in real time, pinpointing where a runner was too aggressive, too conservative, or optimally positioned.

The scoreboard also includes a leaderboard that evaluates players throughout the season. Each runner receives a grade based on how closely their positioning aligns with what the model recommends.

Each scoreboard display includes game information, teams, and highlights the runner involved on the leaderboard to provide context. While we did not factor any of these details into the model's recommendation, they allow the fans to connect the visualization to something that really happened.

Together, these components transform tracking data into an interactive experience that helps audiences evaluate baserunner decisions rather than only judging results.

(3♦) Reconstructing the Moment: Finding the Tell with Tracking Data

To evaluate baserunning decisions fairly, we needed data that captured the game the way the player experienced it, moment by moment. For this project, SMT provided minor league player tracking data, ball tracking data, ball events, and game information from one season of play. To protect privacy, all teams, players, dates, and years were anonymized. Teams are represented by unique three-letter codes, while players are represented by unique four-digit identification numbers. These identifiers have no correlation with official team abbreviations, player numbers, or actual game information. Despite anonymization, the data lets us reconstruct each play from the runner's perspective.

The first step in this process was to connect these anonymized data sources into individual plays. The provided game information consisted of a game string corresponding to every game in the data set. This allows us to connect ball events such as hits and outs with the player and ball tracking data for each play. From there, we filtered the dataset to focus on situations where the decision mattered most: outfield fly balls with only a runner on second base.

For every selected play, we tracked the movements of the runner, the ball, and the fielders throughout the flight of the ball. This allowed us to measure the information available to the runner

at each moment. These measurements lead us to an important question: **Given available information, where should the runner have been?**

The distinction between mid-play decisions and outcomes is what separates our approach from what fans see with traditional baserunning decisions. A player can make a favorable decision and still experience a negative outcome. If we only look at the results, these situations are often impossible to separate. The provided tracking data allows us to evaluate the decisions being made when they happen.

(4♠) Stacking the Odds: Modeling the Optimal Baseball Decision

Understanding the odds is what separates a calculated decision from a guess. A baserunner rarely has complete information when making decisions. They must constantly evaluate the likelihood of different outcomes and choose the action that gives them the best chance of success.

To recreate this decision making process, we developed a model that evaluates a runner's options throughout the flight of a ball. The process began by identifying plays where a meaningful decision exists. We focused on balls hit into the air that traveled beyond the infield and remained in the air long enough for a runner to react. These plays represent situations where a runner must balance competing risks: moving far enough off the base to maximize advancement potential while also being able to safely return on a catch and potentially tag up.

Once we identified these plays, we modeled the uncertainty that comes with the decision. The first challenge was determining how likely a ball was to be caught at any point during its flight. Using ball tracking and fielder movement data, we estimated where and when the ball would become catchable, then evaluated the ability of the fielder to catch the ball using a logistic GAM model. These individual probabilities were then combined in a multinomial logistic regression model to determine the probability of a catch by anyone on the field.

We then modeled the possible outcomes that follow each decision. If the ball is caught, the runner faces the risk of being doubled off or losing the opportunity to successfully tag up. If the ball dropped, the runner can potentially gain extra bases depending on their position and movement. These outcomes were modeled using runner position, runner velocity, the projected landing location of the ball, and expected difficulty of advancing in logistic and ordinal regression models.

The final step combined these models into a decision-making process. At each moment, the model considers possible movements the runner could make, including accelerating forward or retreating toward the base. For each potential action, the model estimates the probability of each outcome and weights those by their expected run value. The movement choice with the highest expected value becomes the model's optimal decision at that moment.

This process creates an ideal baserunner constrained by realistic acceleration and speed. It reveals the decision, the odds behind each option, and the movement with the highest expected value. A more detailed description of the modeling process can be found in Appendix B.

(5♣) The Long Game: Creating a Leaderboard for Baserunning Decisions

One play rarely defines a great baserunner. A player can make the correct decision, position themselves perfectly, and be unsuccessful because of an unpredictable outcome. Over time, however, the best runners separate themselves by consistently maximizing their chances of success.

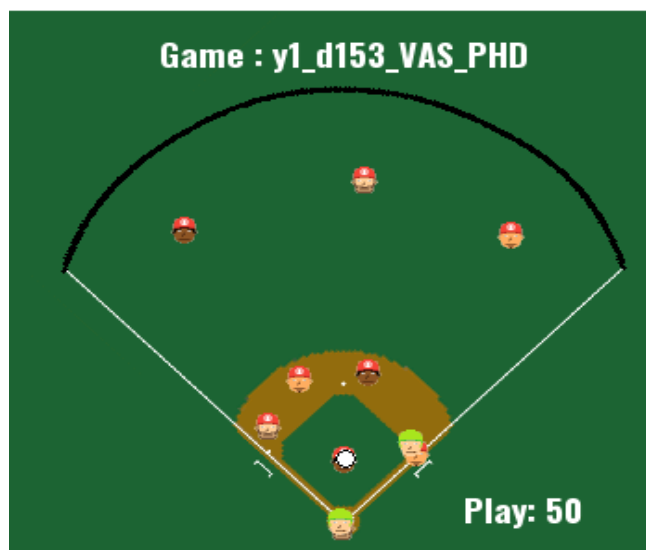
 THE ADVANTAGE TABLE 				
Baserunning leaderboard powered by expected value modeling				
RANK	RUNNER	DECISION SCORE	PLAYS	GRADE
1	PHD-1580	100.09	13	A+
2	ARN-8020	100.02	8	A+
3	PHD-0812	99.97	9	A
4	ANI-7007	99.81	9	A
5	ARN-6201	99.79	7	A-
6	ANI-9818	99.75	7	A-
7	VAS-1500	99.60	5	B+
8	ARN-8108	99.54	8	B+
9	ARN-2168	99.53	6	B+
10	PHD-4573	99.52	10	B+
11	VAS-2605	99.51	5	B+

Figure 3: The Advantage Table ranks baserunners on decision making efficiency across the season. Each row represents a player's average efficiency score, calculated by comparing their actual positioning on fly-ball opportunities with the model's recommended positioning.

To capture this idea, our scoreboard includes a leaderboard that ranks players based on their decisions throughout the season (See **Figure 3**). Rather than looking at traditional statistics like RBIs or stolen bases, the leaderboard measures how closely on average each player's positioning aligns with our recommendation. For each fly-ball, the model compares the runner's movement with the path that would maximize expected value. Runners who consistently position themselves near the optimal path are rewarded, while larger differences create an efficiency gap. This creates a new way for fans to evaluate baserunning. By focusing on positioning, the leaderboard highlights runners who consistently make strong decisions.

(6♥) The Final Turn: Bringing Decision Quality to Baseball Fans

Baseball has always been a game where decisions are made faster than they can be explained. A baserunner must continuously evaluate the ball, the defense, and the situation while making choices that can all change the outcome of the play. Traditional statistics focus primarily on results, leaving it difficult for fans to gauge the quality of decision making.



[Figure 4](#): This play demonstrates a runner whose movement closely matched the model's recommended path, balancing the risk of being doubled off with the opportunity to advance.

Our scoreboard aims to reveal those hidden choices by showing fans the decisions behind each play. By combining player movement with probabilistic modeling, we estimate optimal runner positioning and compare it to the player's path. [Figure 4](#) shows one example from the provided dataset where the runner's movement closely aligns with the model's recommendation, demonstrating how tracking technology can capture the instincts that define quality baserunning.

A necessary consideration in creating a fan-facing analytics tool is understanding how analytics can be interpreted. When a metric or leaderboard is displayed publicly, it can easily shift from a tool for understanding into a source of individual criticism. Because of this, the goal of this project isn't to create a system to judge players, but to provide context behind in-game decisions.

Our scoreboard is designed to encourage curiosity rather than criticism. By visualizing the runner's movement alongside the model's recommendation, fans can appreciate the difficulty that comes with baserunning. The leaderboard follows this same philosophy by emphasizing long-term decision quality rather than focusing on individual mistakes.

Beyond fan engagement, this also creates opportunities for teams to better understand baserunning. Coaches can identify tendencies, provide objective feedback, and better understand where a player is too passive or aggressive.

The best poker players are not remembered because every hand works in their favor, but because they consistently make decisions with favorable odds. Fans should evaluate baserunners through the same lens, recognizing the quality of decisions with the cards players are dealt.

Appendix A: Other Potential Adaptations

While this paper presents a single implementation of the scoreboard, the framework is intentionally flexible. The visualization can be adapted to match a team's branding, audience, or entertainment goals without changing the analytics themselves.

One possible adaptation would be to replace the players in the animation with team mascots or other stylized characters. Rather than directly comparing individual players, a visualization with mascots could communicate the same concepts in a more lighthearted way. This approach may encourage engagement and additionally reduce criticism towards players.

The scoreboard could also become interactive. Before the model recommendation is revealed, fans could be asked to predict the optimal baserunner position or choose between two possible baserunning paths. The visualization would then reveal the model's recommendation, allowing spectators to compare their intuition with an analytic solution.

By separating the analytics from the presentation, organizations can customize the scoreboard to fit their goal, whether that be education, entertainment, or fan interaction. The models we have built can support a variety of viewing experiences.

Appendix B: The Process

B.1 Play Identification and Data Filtering

The analysis was restricted to batted balls satisfying specific criteria designed to isolate plays involving meaningful baserunning decisions. Candidate plays were identified as balls put into play that remained airborne for at least two seconds and traveled beyond the infield. These restrictions excluded routine infield plays and situations where the optimal baserunning strategy is largely predetermined.

The two-second threshold was selected to ensure that the runner had sufficient time to observe the ball trajectory and adjust their positioning. Plays meeting these criteria were subsequently classified based on whether the ball was caught.

B.2 Catch Probability

A primary component of the model was estimating the probability that a ball would be caught at each point during its flight. Since catch probability changes constantly, the model estimated catch probability at each timestamp rather than assigning one probability for each batted ball.

B.2.1 Ball Trajectory Projection

Two reference points along the ball trajectory were identified:

1. The projected point of ground contact
2. The projected point at which the ball descended to eight feet above the playing surface.

These points were selected because they represent meaningful locations for evaluating defensive opportunities. The ground contact represents the final possible catch opportunity, while the eight-foot point represents an earlier point where the fielder could make a play.

Because the tracking dataset did not assume a constant field elevation, a K-Nearest Neighbor (KNN) model was constructed using observed ball-ground interactions across all available fields. This model estimated the playing surface elevation as a function of field location.

For each batted ball, the observed trajectory prior to the final recorded airborne timestamps was used to project the remaining flight path. Generalized Additive Models (GAMs) were fit to estimate the continuation of the trajectory, allowing the projected path to be intersected with the estimated field surface. The resulting intersection provided the predicted time and location of ground contact.

The same procedure was applied to determine the predicted time and location at which the ball reached eight feet above the playing surface.

B.2.2 Fielder-Level Catch Models

For each fielder, variables describing the difficulty of reaching the projected catch location were generated. There variables included:

- Euclidian distance between the fielder and the projected catch location
- Time remaining until the projected catch opportunity
- Fielder velocity toward the projected location
- Directionality of required movement
- Distance of projected catch locations to the outfield wall (wall distances were also estimated from given ball-wall interactions in the data)

A logistic GAM was used to estimate the probability that each individual fielder successfully completed the catch.

Because multiple fielders may have simultaneous opportunities to catch a ball, individual catch probabilities could not simply be summed. Such an approach would result in probabilities exceeding one in situations with multiple fielders in position. Therefore, we used a multinomial logistic regression model to estimate the probability distribution across all outcomes, including each fielder and the outcome of a drop. Catch probabilities of each individual fielder were used to create weighted averages of variables such as throwing velocity and velocity to potentially be used in runner modeling.

B.3 Outcome Classification and Results

The following outcome models required identification of runner results following each play. Since runner outcomes were not directly provided in the dataset, outcomes were inferred using subsequent game states and tracking information.

For plays where future game states were available, runner status was determined by comparing base occupancy on the following play. For example, if a runner attempted to advance from second base and the subsequent play occurred in the same half inning without a runner occupying third base, the runner was deemed unsuccessful.

For cases where future game states could not be reliably used, runner and fielder positioning were used. The distance between the runner and target base, combined with the location of the fielder possessing the ball, was used to estimate whether the runner reached safely.

These observations were used to construct a classification system for determining successful and unsuccessful baserunning outcomes.

B.4 Runner Outcome Models

Initial attempts were made to model runner outcomes at every timestamp during ball flight. However, this approach introduced substantial complexity due to the dependence of advancement outcomes on game state variables such as outs and baserunner situations. Therefore, we based all outcome models on the runner's final observed position immediately prior to either the catch or the ball reaching the ground.

B.4.1 Runner Position Representation

To standardize runner positioning across different starting bases, a continuous basepath variable was developed. This was calculated as

$$\text{Basepath} = \text{Last Base Touched} + \frac{\text{Distance Traveled from Previous Base}}{\text{Distance to Previous Base} + \text{Distance to Next Base}}$$

This transformation allowed runner locations to be represented on a continuous scale. For example, a runner halfway between first and second base would have a basepath value of 1.5. Runner velocity was calculated as the change in basepath over time, measured in basepaths / second:

$$\text{Runner Velocity} = \frac{\Delta \text{Basepath}}{\Delta \text{time}}$$

This simplified both regression modeling and simulation of future runner movement. In animations, basepath is converted back into cartesian coordinates on the field using average coordinates at the modeled runners given basepath.

B.5 Doubled-Off and Tag-Up Models

The probability of a runner being doubled off following a catch and the probability of successfully advancing on a tag up attempt were estimated using logistic regression models.

Predictor variables included:

- Runner basepath position
- Runner basepath velocity
- Distance from the projected landing location to the relevant base

These models provided outcome probabilities for scenarios in which the ball was successfully caught.

B.6 Advancement Models

Advancement probabilities following a ball reaching the ground were modeled separately based on the runner's starting base. For runners beginning on first and second base, ordinal regression models were developed to estimate the number of bases gained. Ordinal regression was selected because advancement outcomes possess an inherent ordering structure. For runners beginning on third base, insufficient observations existed for an advancement model solely based on that data. Because nearly all observed plays resulted in either scoring, a logistic regression model based on all data was used to estimate the probability of advancing at least one base.

Predictor variables included:

- Runner basepath position
- Runner velocity
- Projected ball landing location
- Projected ball landing location distances to base

Additionally, estimated athletic constraints were incorporated into the modeling process. Maximum runner speeds and field throwing velocities were calculated using 95th percentile values among players with sufficient high- effort observations. We also attempted to use weighed-average velocity discussed earlier as projected momentum of the fielder, but did not come back as significant in these models.

B.7 Optimal Runner Simulation

The final model component generated a simulated optimal runner trajectory. The purpose of this simulation was to isolate decision quality from differences in athletic ability. The simulation began several timestamps after contact to incorporate human reaction time. At each timestamp, a range of possible accelerations was evaluated. Each candidate acceleration generated a corresponding projected velocity and position, subject to predefined constraints.

For each candidate movement, two future states were generated:

1. A maximum-retreat state, representing the runner returning toward the original base.
2. A maximum-advance state, representing continued movement toward additional bases.

The maximum-retreat state was evaluated using the doubled-off and tag-up models, while the maximum advance state was evaluated using the advancement models.

The catch probability model was then combined with the conditional outcome models to calculate the probability of each possible outcome:

- Runner doubled off
- Runner remains at original base
- Successful tag-up
- Advancement of zero, one, two, or three bases

Expected run value is calculated as

$$EV = \sum_i P_i \cdot RE_i$$

Where P_i represents the probability of outcome i , and RE_i represents the corresponding run expectancy. [Run expectancies](#) were taken from the 2025 MLB season, as the deviation in relative run expectancies of situations between MLB and MiLB over the past few seasons should be extremely minimal. In actual use, this would obviously be adjusted to said leagues current run expectancies.

For advancement outcomes, ordinal regression estimates on the batter were incorporated to account for their effect on the play. For example, for a runner on first, it becomes more important to try to advance to third if the batter is likely able to reach second if the ball lands. Otherwise, remaining at second would hold up the batter at first.

The acceleration maximizing expected run value was selected as the optimal action at each timestamp. Repeating this optimization process throughout the duration of the play produced the modeled optimal runner trajectory.

We use these run expectancies along with calculated run expectancies for the runners actual path to calculate our efficiency score, which the runners are ranked by on the leaderboard. The score is calculated as the average run expectancy of the runner's actual path divided by the average run expectancy of the model's path.

Appendix C: Limitations

C.1 Data Limitations

Our analysis is based entirely on tracking data from Minor League Baseball. Because these players are still developing, their decision-making may not reflect choices that would be made at the Major League level. Teams often have individual goals for players outside of winning. For example, a pitcher might be asked to cut down on walks and stay in the strike zone more. Differences in player ability, defensive positioning, arm strength, power, and baserunning instincts across all levels can influence the decision runners make. As a result, the model should be interpreted as an evaluation of decision making within the available dataset rather than a universal measure.

C.2 Factors Beyond the Model

The model cannot account for many of the human factors that influence baserunning decisions. Base coaches may have information on an outfielder's arm strength or positioning, runners may trust themselves more than the model, and teams may incorporate scouting reports into their strategy. Additionally, unpredictable events during a play, such as a runner losing their footing or making a poor turn, can affect the outcome despite a sound decision. These contextual factors are difficult to quantify but have an important rule in decision making.

C.3 Game Situation Considerations

Our model assumes that runners seek to maximize expected bases gained while also accounting for the risk of being doubled off. A runner protecting a one-run lead in the ninth inning may behave differently than a runner in a lopsided game. Although the model captures the risk-reward balance, it doesn't incorporate strategic considerations such as score, inning, or game importance.

C.4 Scope of Analysis

This analysis focuses solely on situations in which a single runner is on base while a fly ball is hit. Restricting the analysis to these scenarios simplifies the calculation of run expectancy and isolates the balance between aggressive advancement and avoiding a double play. As tracking technology improves, future work could extend to more complex baserunning situations involving multiple runners and varying game contexts.

Appendix D: Base Dimensions

Base positions were calculated using the updated base dimensions adopted in professional baseball. Although the provided dataset was collected prior to the implementation of larger bases, the updated dimensions were used for consistency with modern field measurements. This adjustment only changes the calculated base locations by approximately an inch or two and is not expected to meaningfully affect the results of our models.